

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing techniques. (BL4,BL5)
Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:40

Annexure A

MOCK INTERVIEW

Code of Conduct:

*I N. Murali Sai (192372309) that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Murali sai

Annexure A

Mock Interview – Sample Questions (5 Questions)

Mock Interview Test Format for Object Oriented Analysis and Design (OOAD)

Section A	Self Introduction & Fundamentals	3 Minutes	5
Section B	Technical Questions (OOAD Concepts)	7 Minutes	10
Section C	UML Modeling & Design	7 Minutes	10
Section D	Case Study / Problem Solving	8 Minutes	10
Section E	HR & Career-Oriented Questions	5 Minutes	5

SECTION A – SELF INTRODUCTION (5 Marks)

Evaluation Criteria

- Communication Skills
- Confidence
- Technical Awareness
- Academic Background

Sample Questions

1. Tell me about yourself.

Object-Oriented Analysis and Design, programming, and UML diagrams. I am a hardworking and confident person, and I always try to improve my technical and communication skills. My goal is to become a good software professional.

2. Why did you choose Computer Science?

I chose Computer Science because I am interested in technology, programming, and problem solving. Computer Science gives me opportunities to create useful software and solve real-world problems. I also enjoy learning new technologies.

3. Explain your favorite subject in OOAD.

My favorite topic in OOAD is UML because UML helps us visually represent a software system. It makes the system easier to understand, design, and communicate with others.

4. What software development tools have you used?

I have used basic programming and development tools such as C programming environments and software development tools used for writing, compiling, and testing programs. I am also learning UML modeling tools.

5. Describe a project where you applied object-oriented concepts.

In a simple software project, I can apply classes and objects to represent real-world entities. For example, in a library system, Book, Student, and Librarian can be represented as classes. Encapsulation can protect data, inheritance can reuse common features, and polymorphism can provide different behaviors.

SECTION B – TECHNICAL QUESTIONS (15 Marks)

Basic Level

Q1. What is Object-Oriented Analysis and Design?

Object-Oriented Analysis and Design, or OOAD, is a method of developing software using objects and classes. Analysis identifies the requirements, objects, and relationships in the problem. Design converts these requirements into a software structure that can be implemented.

Q2. Differentiate between Analysis and Design.

Analysis focuses on understanding the problem and what the system should do. Design focuses on deciding how the system will be built. In simple words, analysis is about “what” and design is about “how”.

Q3. Explain Encapsulation with an example.

Encapsulation means combining data and methods in a class and restricting direct access to the data. For example, a BankAccount class can keep the balance private and provide public methods such as deposit() and withdraw() to access it safely.

Q4. What is Inheritance? Mention its types.

Inheritance is a feature in which one class gets properties and methods from another class. Common types are single, multilevel, hierarchical, multiple, and hybrid inheritance. Inheritance promotes code reuse.

Q5. Define Polymorphism and its benefits.

Polymorphism means “many forms.” It allows the same method or interface to perform different actions depending on the object. Its benefits include flexibility, code reuse, and easier maintenance.

Q6. What is Abstraction?

Abstraction means showing only the important features of an object and hiding unnecessary implementation details. For example, an ATM shows options such as withdraw and deposit without showing the internal banking process.

Q7. Difference between Aggregation and Composition.

Aggregation is a weak “has-a” relationship where the parts can exist independently. Composition is a strong “has-a” relationship where the parts depend on the whole. For example, a Department and Teacher can use aggregation, while a House and Room can be represented using composition.

Q8. What is Coupling and Cohesion?

Coupling is the level of dependency between different classes or modules. Low coupling is preferred. Cohesion is how closely related the responsibilities inside a class or module are. High cohesion is preferred.

Q9. What is a Class and an Object?

A class is a blueprint that defines data and behavior. An object is an instance of a class. For example, Car is a class and a particular car such as myCar is an object.

Q10. Explain Object Responsibility.

Object responsibility means the task or duty assigned to an object. An object should perform the operations related to its data and role. For example, a BankAccount object is responsible for depositing and withdrawing money.

SECTION C – UML MODELING (15 Marks)

Q11. What is UML?

UML stands for Unified Modeling Language. It is a standard visual language used to specify, visualize, design, and document software systems. UML provides diagrams such as use case, class, sequence, activity, state, component, and deployment diagrams.

Q12. When would you use a Use Case Diagram?

A Use Case Diagram is used during requirements analysis to show the interaction between users or external systems and the system. It identifies actors and the functions, or use cases, provided by the system.

Q13. Draw and explain a Class Diagram for a Library Management System.

A simple Library Management System can contain the classes Book, Member, Librarian, and Library.

Book: bookId, title, author; operations: issueBook(), returnBook().

Member: memberId, name; operations: borrowBook(), returnBook().

Librarian: librarianId, name; operations: addBook(), removeBook().

Library: libraryId, name; operations: searchBook(), manageBooks().

Relationships: Members borrow Books, Librarian manages Books, and Library contains Books and Members.

Q14. Differentiate Sequence Diagram and Activity Diagram.

A Sequence Diagram shows the order of messages exchanged between objects over time. An Activity Diagram shows the workflow or sequence of activities in a process. Sequence diagrams focus on object interaction, while activity diagrams focus on process flow.

Q15. Explain State Diagram with an example.

A State Diagram shows the different states of an object and the transitions between those states. For example, an Order can have states such as New, Confirmed, Shipped, Delivered, and Cancelled.

Q16. What is a Deployment Diagram?

A Deployment Diagram shows the physical arrangement of hardware nodes and software components in a system. It helps explain where software is deployed and how different nodes communicate.

Q17. How do Component Diagrams help software architects?

Component Diagrams show the major software components and their dependencies. They help architects understand the system structure, interfaces, dependencies, and how components communicate.

Q18. What is Multiplicity in UML?

Multiplicity specifies how many objects of one class can be associated with an object of another class. Examples are 1, 0..1, 1..*, and *. For example, one customer can place many orders, represented as 1 to many.

SECTION D – CASE STUDY / PROBLEM SOLVING (10 Marks)

Case Study 1

A hospital wants a system to manage:

- Patients
- Doctors
- Appointments
- Billing

Questions

Case Study 1 – Hospital Management System

1. Classes: Patient, Doctor, Appointment, and Billing.
2. Simple relationships: Patient books Appointment; Doctor attends Appointment; Appointment generates Billing.
3. Relationships: Patient–Appointment, Doctor–Appointment, and Appointment–Billing.
4. Useful UML diagrams: Use Case Diagram, Class Diagram, Sequence Diagram, and Activity Diagram.
5. Inheritance: A general Person class can be created, with Patient and Doctor as specialized classes. Common attributes such as name and contact can be reused.

Case Study 2

An Online Shopping System supports:

- Customers
- Products
- Cart
- Orders
- Payments

Questions

1. Actors: Customer, Admin, and Payment Service. Main use cases include browse products, add to cart, place order, make payment, and manage products.
2. Main classes: Customer, Product, Cart, Order, and Payment.
3. Suitable design patterns: Factory Pattern can be used to create payment objects; Strategy Pattern can be used for different payment methods.
4. Polymorphism in payment processing: A common Payment interface can define pay(). Classes such as CardPayment, UpiPayment, and CashOnDelivery can implement pay() differently. The system can call pay() without depending on the specific payment type.

SECTION E – HR & PROFESSIONAL QUESTIONS (5 Marks)

Q19. Why should we hire you?

You should hire me because I am hardworking, willing to learn, and interested in technology. I have a positive attitude and I can work well with others. I will continuously improve my technical skills and contribute to the organization.

Q20. What are your strengths and weaknesses?

My strengths are that I am hardworking, responsible, and willing to learn new things. One weakness is that I sometimes spend extra time making sure my work is correct. I am improving this by planning my time and setting priorities.

Q21. How do you handle project deadlines?

I handle deadlines by first understanding the requirements and dividing the work into smaller tasks. I set priorities, follow a schedule, and regularly check my progress. If I face a problem, I ask for help early so that the deadline is not affected.

Q22. Have you worked in a team project? Explain your role.

Yes. In a team project, I can contribute by completing my assigned tasks, sharing ideas, communicating with team members, and helping to test and improve the project. I believe teamwork and communication are important for successful project completion.

Q23. Where do you see yourself in five years?

In five years, I see myself as a skilled software professional with strong knowledge of software development and object-oriented design. I want to work on real-world projects, take more responsibilities, and continue learning new technologies.

RUBRICS FOR CALCULATION:
MOCK INTERVIEW

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately